

Coding & Solution

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Mcharan07/A_Comprehensive_Measure_of_Well-Being/
Programming Language(s)	Python 3.12
Framework(s) Used	Flask 3.0.3, scikit-learn 1.5, Pandas 2.2, NumPy 1.26, Jupyter Notebook
Key Features Implemented	• Linear Regression model trained on 167-country HDI dataset • Flask web app: Home, Predict, About, 404 pages • Input validation with error banners • Four-tier HDI classification (Low / Medium / High / Very High) • Tier gauge visual on predict page • REST API endpoint /api/predict (JSON in / JSON out) • Full EDA notebook (heatmap, strip plot, scatter, pairplot, residual) • Model persisted to HDI.pkl via Pickle
Pending / Incomplete Features	• User authentication (not in scope for ML-0027) • Database storage of prediction history • Multi-country comparison view
Setup / Run Instructions	Training: cd Training && pip install -r requirements.txt && jupyter notebook HumDevIndex.ipynb Flask: cd Flask && pip install -r requirements.txt && python app.py Open: http://127.0.0.1:5000/

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes